Algebra 2

SPECIAL PRODUCTS

Expressions where the values can be obtained without execution of long multiplication.

With x, y and z as real numbers or variables or algebraic expression, the following are the special products.

1. Sum and difference of same terms or Difference of two squares

$$(x+y)(x-y)=x^2-y^2$$

2. Square of a binomial

$$(x+y)^2=x^2+2xy+y^2$$

$$(x-y)^2=x^2-2xy+y^2$$

3. Cube of a binomial

$$(x+y)^3=x^3+3x^2y+3xy^2+y^3$$

$$(x-y)^3=x^3-3x^2y+3xy^2-y^3$$

4. Difference of two cubes

$$x^3-y^3=(x-y)\left(x^2+xy+y^2\right)$$

5. Sum of two cubes

$$x^3+y^3=(x-y)\left(x^2-xy+y^2\right)$$

6. Square of a trinomial

$$(x+y+z)^2=x^2+y^2+z^2+2xy+2xz+2yz$$

REMAINDER THEOREM AND FACTOR THEOREM

Remainder Theorem states that if a polynomial in an unknown quantity x is divided by a first degree expression in the same variable, (x – k), where k may be any real number or complex number, the remainder to be expected will be equal to the sum obtained when the numerical value of k is substituted for x in the polynomial. Thus,

$$\text{remainder}=\underset{x\to k}{f(x)}$$

Factor theorem states that if a polynomial is divided by (x – k) will result to a remainder of zero, then the value (x – k) is a factor of the polynomial.

Both remainder theorem and factor theorem were suggested by a French mathematician, Etienne Bezout (1730 – 1783).

What is an **Exponent**?

Exponent is a number that gives the power to which a base is raised. For example, in 3^2, the base is 3 and the exponent is 2.

Exponent should not be misunderstood as "power" Power is a word that is almost never used in its correct, original sense any more. Strictly speaking, if we write 32 = 9, then 3 is the base, 2 is the exponent and 9 is the power. But almost everyone, including most mathematicians, would say that 3 is the power and that "power" and "**exponent**" mean the same thing. The misuse has probably come from a misunderstanding of statements such "nine is the second power of three".

Properties of **Exponent**

Property	Definition	Example
Property of 1	$a^1=a$	$2^1=2$
Property of 0	$a^0=1$	$5^0=1$
Property of negatives	$a^{-m}=\frac{1}{a^m}$	$4^{-2}=\frac{1}{4^2}=\frac{1}{16}$
Product property	$a^m\cdot a^n=a^{m+n}$	$2^2+2^3=2^{2+3}=2^5=32$
Quotient property	$\frac{a^m}{a^n}=a^{m-n}$	$\frac{2^3}{2^2}=2^{3-2}=2^1=2$
Power of a product	$(ab)^m=a^mb^m$	$(2x)^4=2^4x^4=16x^4$
Power of a quotient	$\left(\frac{a}{b}\right)^m=\frac{a^m}{b^m}$	$\left(\frac{2}{x}\right)^4=\frac{2^4}{x^4}=\frac{16}{x^4}$
Power of a power	$\left(a^m\right)^n=a^{mn}$	$\left(y^6\right)^2=y^{12}$
Property of fraction	$a^{\frac{m}{n}}=\sqrt[n]{a^m}$	$(4x)^{\frac{5}{3}}=\sqrt[3]{(4x)^5}$

What is **Logarithm**?

The **logarithm** of a number or variable x to base b, $\log_b x$, is the exponent of b needed to give x.

$\text{Log}_2 16 = 4$ may be written as $2^4 = 16$

Properties of **Exponent**

Property	Definition	Example
Product rule	$\log(xy)=\log x+\log y$	$\log(2x)=\log 2+\log x$
Quotient rule	$\log\left(\frac{x}{y}\right)=\log x-\log y$	$\log\left(\frac{5}{2}\right)=\log 5-\log 2$
Power rule	$\log x^n=n\log x$	$\log 5^3=3\log 5$
Change of base rule	$\log_a x=\frac{\log_b x}{\log_b a}$	$\log_5 25=\frac{\log_b 25}{\log_b 5}$
Equality rule	If $\log x=\log y$, then $x=y$	$\log 3x=\log 5y\Rightarrow 3x=5y$
Equal base and variable	$\log_a a=1$	$\log_5 5=1$

The term "logarithm" comes from Greek words, "logus" meaning "ratio" and "arithmus" meaning "number". John Napier (1550 – 1617) invented logarithm in 1614 using e = 2.718… for its base. Logarithm with base e (loge or ln) is called the **natural logarithm** or the **Napierean logarithm**. In 1616, through the suggestion of John Napier, Henry Briggs (1561 0 1630), a professor of Geometry at Gresham College in London, improved the logarithm using 10 as the base. The logarithm with base 10 is known as **common logarithm** or the **Briggsian logarithm**.

Relation between Natural & Common Logarithms

The natural logarithm can be converted into a common logarithm and vice versa. To obtain this, a factor known as the modulus of logarithm is necessary, such as:

$$\log x=0.4343\ln x \qquad \ln x=2.3026\log x$$

The coefficients 0.4343 and 2.3026 are the referred to as the **modulus of logarithm**.

LCD, LCM & GCF

Least common denominator (LCD) refers to the product of several prime numbers occurring in the denominators, each taken with its greatest multiplicity.

Problem:
What is the least common denominator of 8, 9, 12 and 15?

Solution:

$8=2^3 \qquad 12=3\cdot 2^2 \qquad \text{LCD}=2^3\left(3^2\right)(5)$

$9=3^2 \qquad 15=3\cdot 5 \qquad \text{LCD}=360$

A common multiple is a number that two other numbers will divide into evenly. The **least common multiple (LCM)** is the lowest multiple of two numbers.

Problem:
What is the least common multiple of 15 and 18?

Solution:

$15=3\cdot 5 \qquad \text{LCM}=3^2(5)(2)$

$18=3^2\cdot 2 \qquad \text{LCM}=90$

A factor is a number that divides into a larger number evenly. The **greatest common factor (GCF)** is the largest number that divides into two or more numbers evenly.

Problem:
What is the greatest common factor of 70 and 112?

Solution:

$70=2\cdot 5\cdot 7$

$112=2^4\cdot 7$

Common factors are 2 and 7.

$\text{GCF}=2(7)$

$\text{GCF}=14$